Empirical & Molecular Formula

Exam code: 0620 & 0971

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Calculating empirical & molecular formulae

Extended tier only

- As previously discussed in Empirical Formulae & Formulae of Ionic Compounds:
- The **empirical formula** is the simplest whole number ratio of the atoms of each element present in one molecule or formula unit of the compound
  - E.g. the empirical formula of ethanoic acid is CH_2O

How to calculate empirical formulae

- Empirical formula calculations are very methodical
- Use a table and the following steps to complete an empirical formula calculation:
  1. Write the element
  2. Write the value given for each element
     - This may be given as a mass, in g, or as a percentage
     - There are exam questions where you are required to calculate the value of one of the elements
  3. Write the relative atomic mass of each element
  4. Calculate the moles of each element
     - Moles = $\frac{\text{mass}}{A_r}$

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Got it
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- If you get a ratio that does not have whole numbers, you multiply by an appropriate number to make all the values into whole numbers

6. Write the final empirical formula

Worked Example

A sample of a compound was found to contain 10 g of hydrogen and 80 g of oxygen.

Calculate the empirical formula of this compound.

$$A_r(H) = 1 \quad A_r(O) = 16$$

Answer:

1. Element	H	O
2. Value	10	80
3. Relative atomic mass	1	16
4. Moles = $\frac{\text{mass}}{A_r}$	$\frac{10}{1} = 10$	$\frac{80}{16} = 5$
5. Ratio (divide by smallest)	$\frac{10}{5} = 2$	$\frac{5}{5} = 1$
6. Answer	The empirical formula is H_2O	

Worked Example

Carbohydrate **X** was analysed and found to contain 31.58% carbon and 5.26% hydrogen by mass.

Find the empirical formula of carbohydrate **X**.

$$A_r(H) = 1 \quad A_r(C) = 12 \quad A_r(O) = 16$$

Answer:

A carbohydrate contains carbon, hydrogen and oxygen

The percentages do not add up to 100%, which means that you need to calculate the percentage of oxygen needs to be calculated

Percentage of oxygen = 100 – 31.58 – 5.26 = 63.16%

1. Element	C	H	O
2. Value	31.58	5.26	63.16
3. Relative atomic mass	12	1	16

5. Ratio (divide by smallest)	$\frac{2.63}{2.63} = 1$	$\frac{5.26}{2.63} = 2$	$\frac{3.95}{2.63} = 1.5$
5. Whole number ratio	1 x 2 = 2	2 x 2 = 4	1.5 x 2 = 3
6. Answer	The empirical formula is $C_2H_4O_3$		

Examiner Tips and Tricks

The molar ratio must be a whole number.

If you don't get a whole number when calculating the ratio of atoms in an empirical formula, such as 1.5, multiply that and the other ratios to achieve whole numbers.

How to calculate molecular formula

- Molecular formula gives the **actual numbers of atoms of each element** present in the formula of the compound

Table showing the relationship between empirical and molecular formulae

Compound	Empirical formula	Molecular formula
Methane	CH_4	CH_4
Ethane	CH_3	C_2H_6
Ethene	CH_2	C_2H_4
Benzene	CH	C_6H_6

- To calculate the molecular formula:
  1. Find the relative formula mass of the empirical formula
     - Add the relative atomic masses of all the atoms in the empirical formula
  2. Use the following equation:
     - $\frac{\text{relative formula mass of molecular formula}}{\text{relative formula mass of empirical formula}}$
  3. Multiply the number of each element present in the empirical formula by the number from step 2 to find the molecular formula

Worked Example

Calculate the molecular formula of **X**.

$$A_r(C) = 12 \quad A_r(H) = 1 \quad A_r(S) = 32$$

Answer:

1. Calculate the relative formula mass of the empirical formula:
   - $M_r = (12 \times 4) + (1 \times 10) + (32 \times 1) = 90$
2. Divide the relative formula mass of **X** by the relative formula mass of empirical formula:
   - 180 / 90 = 2
3. Multiply for the molecular formula:
   - The number of atoms of each elements should be multiplied by 2
   - $(C_{4 \times 2}) + (H_{10 \times 2}) + (S_{1 \times 2})$
   - Molecular formula of **X** = $C_8H_{20}S_2$

Deducing formulae of hydrated salts

- A hydrated salt is a crystallised salt that contains water molecules as part of its structure
- The formula of a hydrated salt shows the water molecules, e.g. $CuSO_4 \bullet 2H_2O$
  - The • symbol shows that the water present is water of crystallisation
- The formula of hydrated salts can be determined experimentally by:
  - Weighing a sample of the hydrated salt
  - Heating it until the water of crystallisation has been driven off
    - This is achieved by heating until a constant mass
  - Re-weighing the anhydrous salt
- From the results, you can determine the mass of anhydrous salt and the mass of the water of crystallisation
- Applying a similar approach to deducing empirical formulae, the formula of the hydrated salt can be calculated

How to calculate water of crystallisation

- The steps for empirical formula can be adapted for hydrated salt / water of crystallisation calculations
  - Instead of writing elements, write the two components of a hydrated salt
    - The salt
    - Water
  - Instead of writing relative atomic mass, write the relative molecular / formula mass of the salt and water
- Use a table and the following steps to complete the calculation:
  1. Write the salt and water

3. Write the relative molecular / formula mass of the salt and water
4. Calculate the moles of the salt and water
   - Moles = $\frac{\text{mass}}{M_r}$
5. Calculate the ratio salt : water
   - Divide all the moles by the smallest number of moles
   - The calculation should give a ratio of **1** salt : ***x*** water
6. Write the final hydrated salt formula

Worked Example

11.25 g of hydrated copper sulfate, $CuSO_4.xH_2O$, is heated until it loses all of its water of crystallisation.

It is re-weighed and its mass is 7.19 g.

Calculate the formula of the hydrated copper(II) sulfate.

A_r (Cu) = 63.5 A_r (S) = 32 A_r (O) = 16 A_r (H) = 1

Answer:

1. Salt and water	$CuSO_4$	H_2O
2. Value	7.19	11.25 - 7.19 = 4.06
3. M_r	63.5 + 32 + (16 x 4) = 159.5	(1 x 2) + 16 = 18
4. Moles = $\frac{\text{mass}}{M_r}$	$\frac{7.19}{159.5}$ = 0.045	$\frac{4.06}{18}$ = 0.226
5. Salt : water ratio	$\frac{0.045}{0.045}$ = 1	$\frac{0.226}{0.045}$ = 5
6. Formula of hydrated salt	The formula is $CuSO_4$•**5**H_2O	

Examiner Tips and Tricks

The specification is not clear about whether deducing the formula of hydrated salts is required.

However, it is an application of deducing empirical formulae so it is worth knowing how to do this.

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